

CCNA project

“Enterprise Network Infrastructure Design Using VLANs, DHCP, EIGRP, and ACLs”



By:

Abdelrahman Khaeld Elsayed

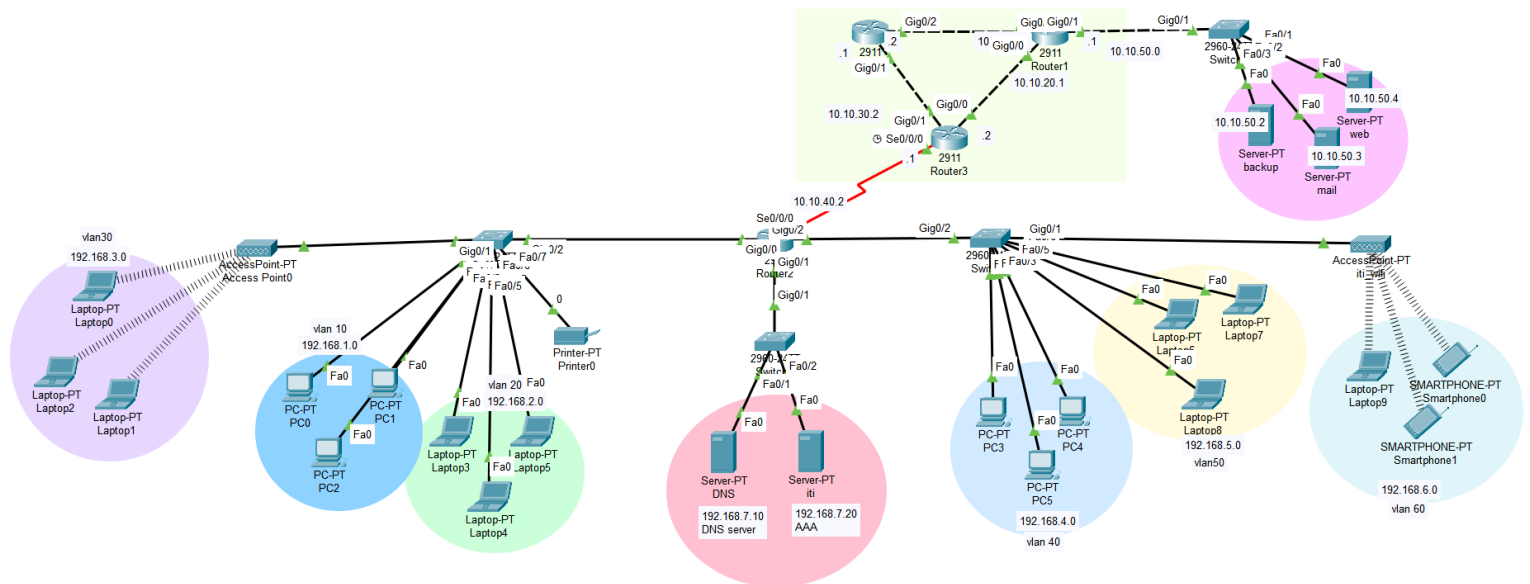
Under Supervision:

Eng. Saleh Mahmoud

1. Introduction

This project presents the design, implementation, and configuration of a complete enterprise network based on CCNA concepts. The network topology simulates a real-world organizational environment using multiple routers, switches, VLANs, servers, and end devices.

This project integrates core CCNA topics including VLANs, trunking, DHCP, dynamic routing, STP, SSH, AAA, and ACLs into a single functional network. It serves as a comprehensive hands-on practice project that reflects real enterprise networking scenarios and reinforces both configuration and troubleshooting skills.



2. Project Objectives

1. Design a scalable enterprise network topology using routers, switches, servers, wireless access points, and multiple end devices.
2. Implement VLAN segmentation to logically separate network departments, reduce broadcast traffic, and enhance network performance and security.
3. Configure inter-VLAN routing to enable communication between different VLANs while maintaining proper network isolation.
4. Deploy DHCP services to automatically distribute IP addresses, subnet masks, default gateways, and DNS server information to all end devices.
5. Implement dynamic routing using EIGRP to ensure efficient route exchange and reliable connectivity between routers.
6. Enable Rapid Spanning Tree Protocol (RSTP) on switches to prevent switching loops and ensure fast convergence.
7. Secure device management access by configuring hostnames, encrypted passwords, SSH-only remote access, and centralized authentication using a RADIUS server.
8. Apply Access Control Lists (ACLs) to control access to network services such as DNS, web, mail, and FTP/TFTP services based on host, network, and VLAN security policies.
9. Configure TFTP services to support network device configuration backup and restoration, simulating real enterprise network maintenance procedures.
10. Integrate enterprise network services including DNS, web, mail, DHCP, AAA (RADIUS), and TFTP to create a complete and functional network environment.

3. Basic Router and server Configuration and Secure Management

The router hostname:

- **Hostname:** Abdo Khaled Elsayed
- **Domain Name:** ITI.com

Password and Security Configuration:

- **Enable Secret:** Configured and encrypted
- **Service Password Encryption:** Enabled

Console Access Configuration:

The console line is secured with a password to prevent unauthorized local access.

- **Console Line:** line console 0
- **Authentication:** Password-based login

Remote Access Configuration (VTY Lines)

Virtual Terminal (VTY) lines are configured to allow secure remote management.

- **VTY Lines:** 0–15
- **Authentication Method:** Local username database
- **Allowed Protocol:** SSH only
- **RSA Key Generation:** Enabled
- **Remote Access Protocol:** SSH

Local User Account Configuration

A local user account is created to support SSH authentication. This user account is used for secure remote login to the router.

- **Username:** Elsayed
- **Password:** 123

```
line con 0
password 7 08204E4A021104121E18
login
!
line aux 0
!
line vty 0 4
password 7 08204E4A021104121E18
login local
transport input ssh
!
!
!
end

Abdo#
```

4. VLANs Table

VLAN ID	Department Name	IP Subnet Range	Default Gateway	Assigned Switch Ports
10	IT	192.168.1.0/24	192.168.1.1	Fa0/1, Fa0/2, Fa0/3
20	Accounting	192.168.2.0/24	192.168.2.1	Fa0/4, Fa0/5, Fa0/6
30	HR	192.168.3.0/24	192.168.3.1	Gig0/1
40	Finance	192.168.4.0/24	192.168.4.1	Fa0/1, Fa0/2, Fa0/3
50	Engineering	192.168.5.0/24	192.168.5.1	Fa0/4, Fa0/5, Fa0/6
60	Users	192.168.6.0/24	192.168.6.1	Gig0/1

```
Abdo#show vlan
```

```

VLAN Name                Status    Ports
-----
1    default                active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24
10   IT                      active    Fa0/1, Fa0/2, Fa0/3
20   Accounting             active    Fa0/4, Fa0/5, Fa0/6
30   HR                      active    Gig0/1

```

```
Elsayed#show vlan brief
```

```

VLAN Name                Status    Ports
-----
1    default                active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24
40   Finance                active    Fa0/1, Fa0/2, Fa0/3
50   Engineering             active    Fa0/4, Fa0/5, Fa0/6
60   users                   active    Gig0/1

```

5. Access point

Two VLANs are extended through wireless access points to provide Wi-Fi access for different departments. Each wireless network is mapped to a specific VLAN, allowing wireless devices to obtain network access according to their assigned department.

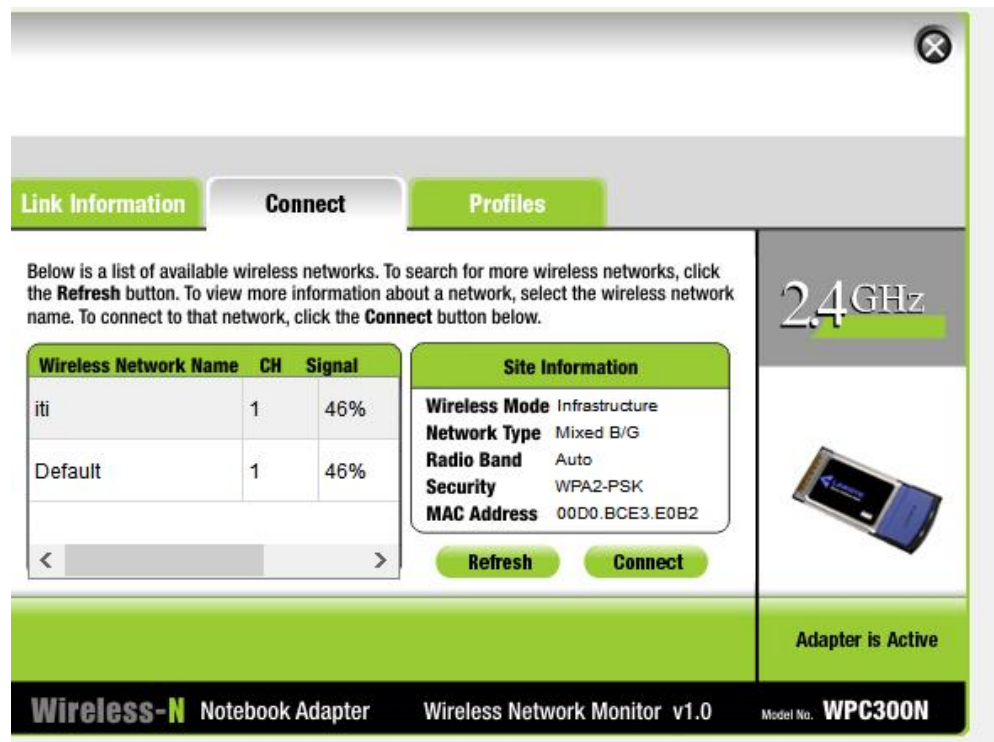
Connected Wireless Devices:

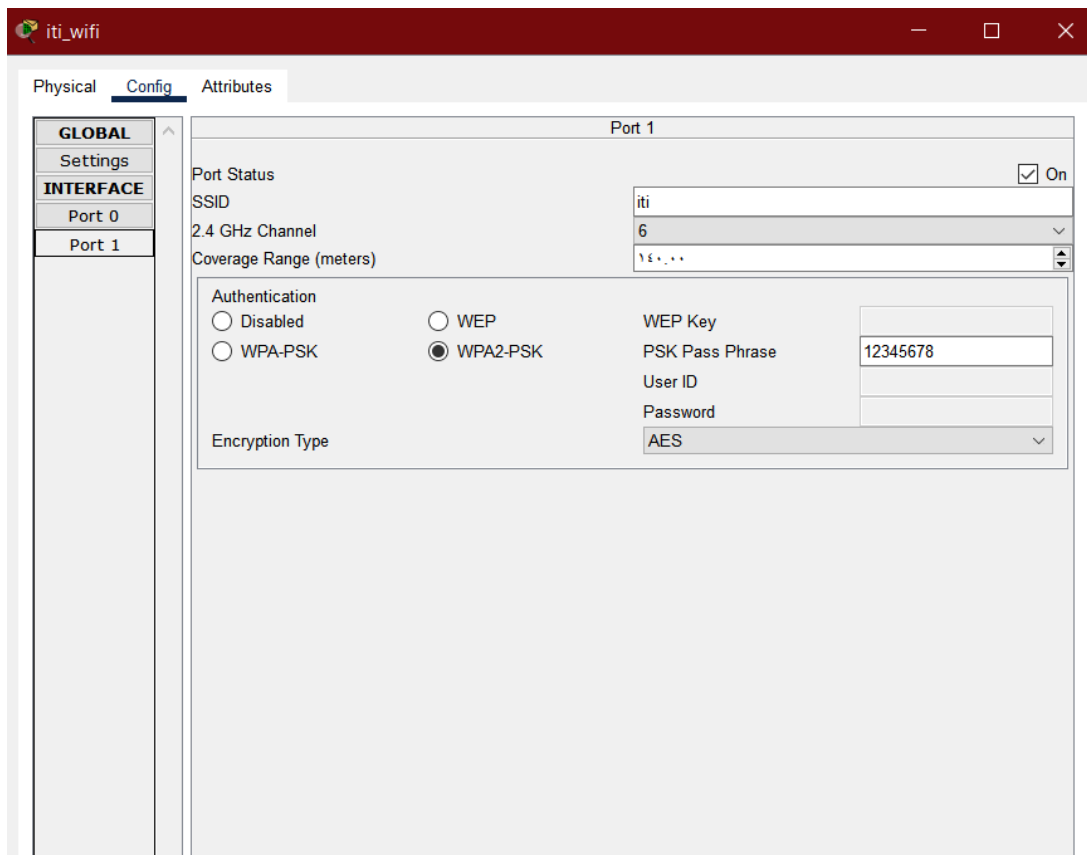
VLAN 30 (Default Wireless Network):

- VLAN ID: 30
- Department: Human Resources
- 3 Laptops
- Purpose: Provides wireless access for HR department users with controlled network access.

VLAN 60 (Wireless Network: ITI):

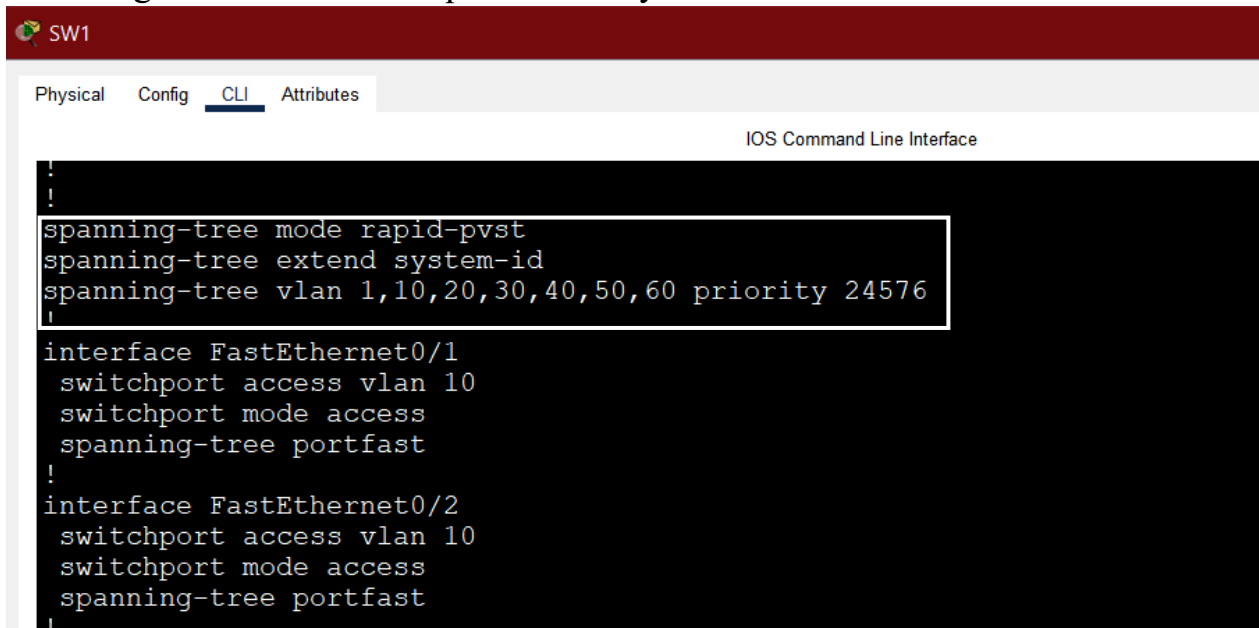
- Department: Users
- SSID Name: ITI
- 1 Laptop
- 2 Smartphones
- Purpose: Provides wireless access for general users and mobile devices.





6. Rapid Spanning Tree Protocol (RSTP)

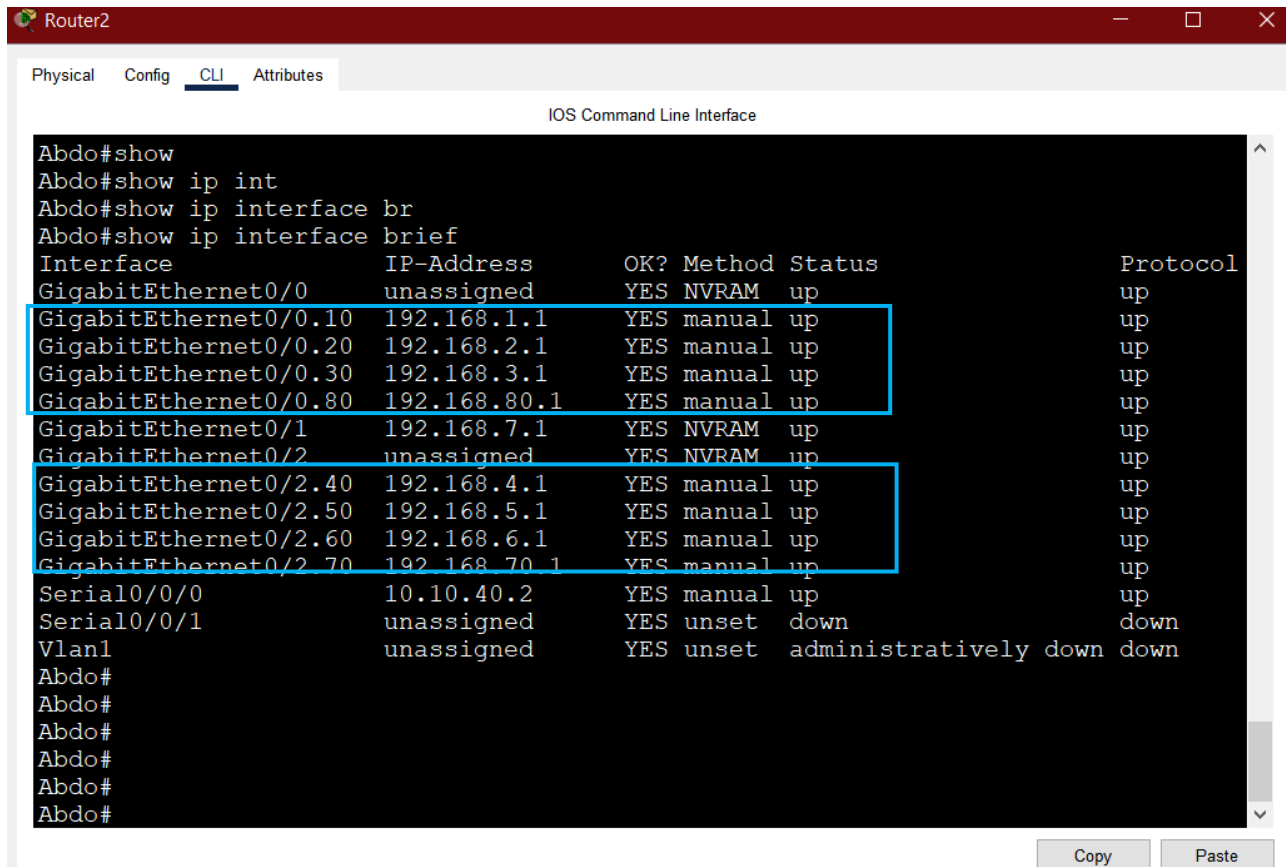
Rapid Spanning Tree Protocol (RSTP) is enabled on all switches to prevent Layer 2 loops and provide fast network convergence. This ensures a stable, loop-free switching environment with quick recovery from link failures.



7. VLAN and Router-On-A-Stick (ROAS) Implementation

Router-on-a-Stick is used in this project to enable inter-VLAN routing using a single physical router interface connected to the switch via an IEEE 802.1Q trunk.

In this design, the switch interface connected to the router is configured as a trunk port. On the router, sub interfaces are created on the physical interface, with each sub interface assigned to a specific VLAN and configured with its own IP address to act as the default gateway for that VLAN.



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Abdo#show
Abdo#show ip int
Abdo#show ip interface br
Abdo#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES NVRAM up up
GigabitEthernet0/0.10 192.168.1.1 YES manual up up
GigabitEthernet0/0.20 192.168.2.1 YES manual up up
GigabitEthernet0/0.30 192.168.3.1 YES manual up up
GigabitEthernet0/0.80 192.168.80.1 YES manual up up
GigabitEthernet0/1 192.168.7.1 YES NVRAM up up
GigabitEthernet0/2 unassigned YES NVRAM up up
GigabitEthernet0/2.40 192.168.4.1 YES manual up up
GigabitEthernet0/2.50 192.168.5.1 YES manual up up
GigabitEthernet0/2.60 192.168.6.1 YES manual up up
GigabitEthernet0/2.70 192.168.70.1 YES manual up up
Serial0/0/0 10.10.40.2 YES manual up up
Serial0/0/1 unassigned YES unset down down
Vlan1 unassigned YES unset administratively down down
Abdo#
Abdo#
Abdo#
Abdo#
Abdo#
Abdo#
```

8. Routing

Routing is implemented in this project to enable communication between different VLANs, network segments, and routers. Both Static Routing and Dynamic Routing (EIGRP) are used

Default routing is configured on selected routers to define a gateway of last resort. This allows routers to forward packets destined for unknown networks to a predefined next-hop router.

The screenshot displays a network simulation environment. On the left, a topology diagram shows a central router (R1) connected to a switch (S1) and a server (Server-PT DNS). The server is labeled with IP 192.168.7.10 and 'DNS server'. Other devices include a printer (Printer-PT) and a laptop (Laptop). The bottom of the interface has a 'Scenario 0' dropdown, 'New' and 'Delete' buttons, and a 'Toggle PDU List Window' button.

On the right, a 'CLI' window titled 'IOS Command Line Interface' shows the following output:

```
IOS Command Line Interface

1 - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.40.0/24 is directly connected, Serial0/0/0
L    10.10.40.2/32 is directly connected, Serial0/0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.2.1/32 is directly connected, GigabitEthernet0/0.20
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.3.1/32 is directly connected, GigabitEthernet0/0.30
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/2.40
L    192.168.4.1/32 is directly connected, GigabitEthernet0/2.40
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, GigabitEthernet0/2.50
L    192.168.5.1/32 is directly connected, GigabitEthernet0/2.50
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/24 is directly connected, GigabitEthernet0/2.60
L    192.168.6.1/32 is directly connected, GigabitEthernet0/2.60
192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.7.0/24 is directly connected, GigabitEthernet0/1
L    192.168.7.1/32 is directly connected, GigabitEthernet0/1
192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.70.0/24 is directly connected, GigabitEthernet0/2.70
L    192.168.70.1/32 is directly connected, GigabitEthernet0/2.70
192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.80.0/24 is directly connected, GigabitEthernet0/0.80
L    192.168.80.1/32 is directly connected, GigabitEthernet0/0.80
S*  0.0.0.0/0 is directly connected, Serial0/0/0
```

Enhanced Interior Gateway Routing Protocol (EIGRP) is deployed as the dynamic routing protocol between routers in the network. EIGRP enables routers to automatically exchange routing information and dynamically update their routing tables when network changes occur.

Physical Config CLI Attributes

IOS Command Line Interface

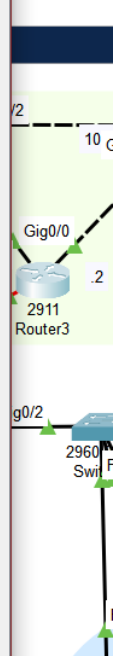
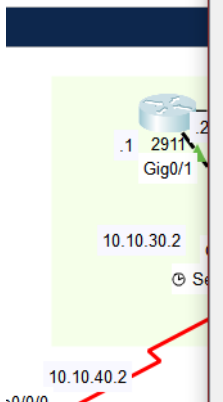
```
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D    10.10.10.0/24 [90/3072] via 10.10.20.1, 00:06:06,
GigabitEthernet0/0
    [90/3072] via 10.10.30.1, 00:06:06,
GigabitEthernet0/1
C    10.10.20.0/24 is directly connected, GigabitEthernet0/0
L    10.10.20.2/32 is directly connected, GigabitEthernet0/0
C    10.10.30.0/24 is directly connected, GigabitEthernet0/1
L    10.10.30.2/32 is directly connected, GigabitEthernet0/1
C    10.10.40.0/24 is directly connected, Serial0/0/0
L    10.10.40.1/32 is directly connected, Serial0/0/0
D    10.10.50.0/24 [90/3072] via 10.10.20.1, 00:06:06,
GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/0

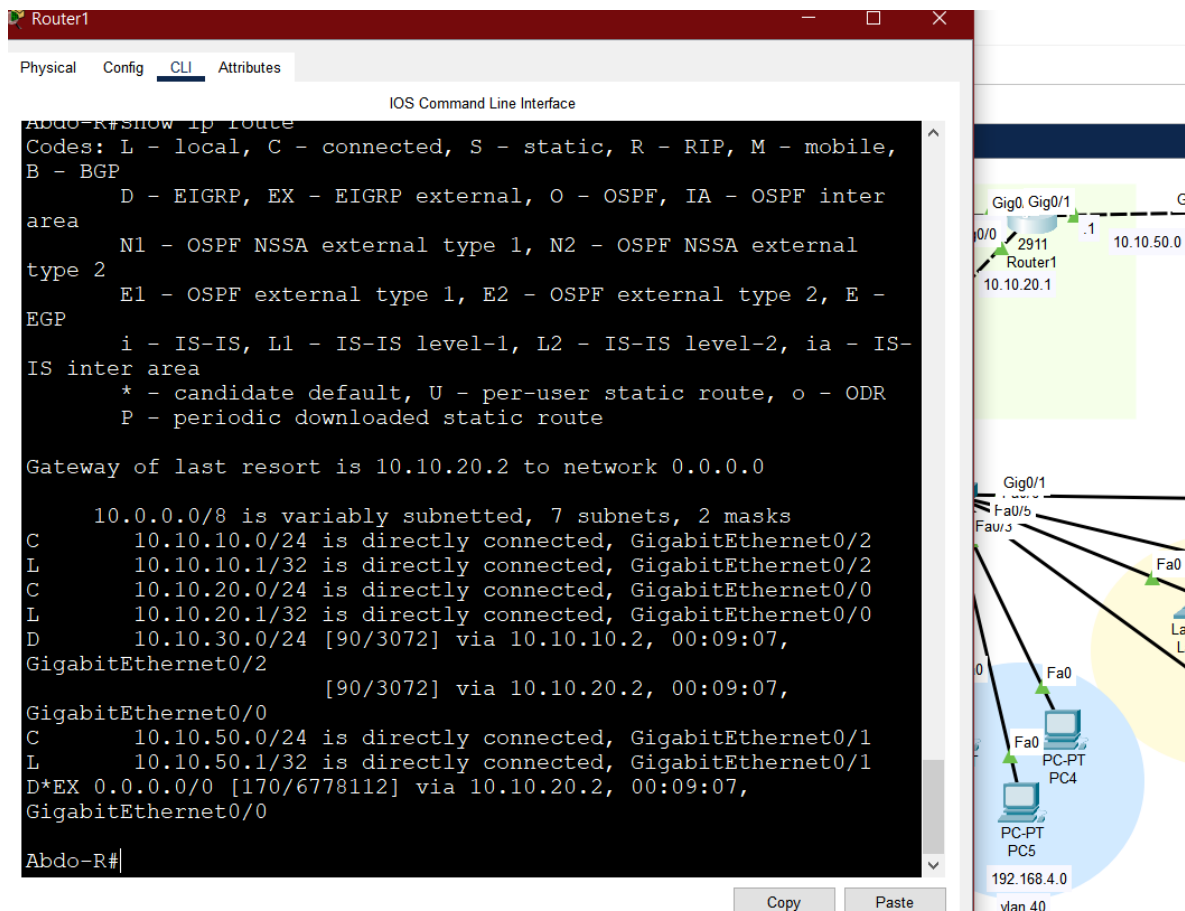
Abdo R3#
```

Conv Paste

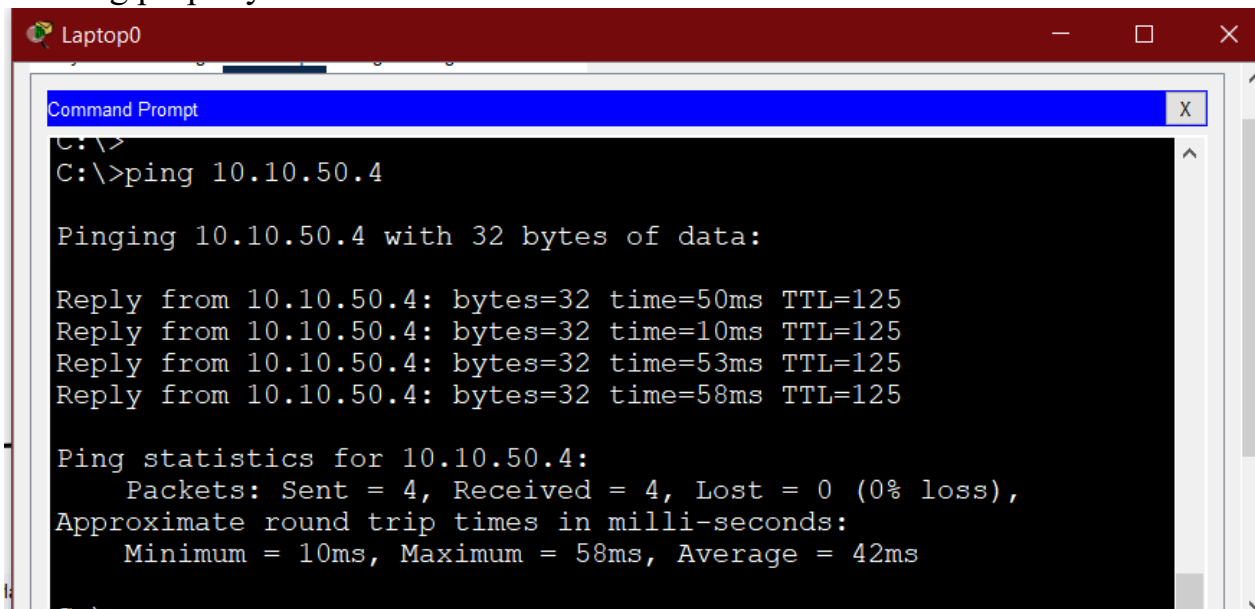



```
Gateway of last resort is 10.10.30.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.10.10.0/24 is directly connected, GigabitEthernet0/2
L    10.10.10.2/32 is directly connected, GigabitEthernet0/2
D    10.10.20.0/24 [90/3072] via 10.10.30.2, 00:07:23,
GigabitEthernet0/1
    [90/3072] via 10.10.10.1, 00:07:23,
GigabitEthernet0/2
C    10.10.30.0/24 is directly connected, GigabitEthernet0/1
L    10.10.30.1/32 is directly connected, GigabitEthernet0/1
D    10.10.50.0/24 [90/3072] via 10.10.10.1, 00:07:23,
GigabitEthernet0/2
D*EX 0.0.0.0/0 [170/6778112] via 10.10.30.2, 00:07:23, GigabitEthe
```



Routing was verified by performing a ping test from VLAN 30 to the remote network 10.10.50.4. The ping was successful, confirming that default static routing and EIGRP are correctly configured and that inter-network communication is working properly.



9. DHCP Configuration

Dynamic Host Configuration Protocol (DHCP) is implemented on the router to automatically assign IP addresses to all end devices in the network. The DHCP service provides IP address, subnet mask, default gateway, and DNS server information, reducing manual configuration and administrative overhead.

Pool Name	Network / Range	Subnet Mask	Leased IPs
IT (VLAN 10)	192.168.1.0 – 192.168.1.254	/24	3
Accounting (VLAN 20)	192.168.2.0 – 192.168.2.254	/24	3
HR (VLAN 30)	192.168.3.0 – 192.168.3.254	/24	3
Finance (VLAN 40)	192.168.4.0 – 192.168.4.254	/24	3
Engineering (VLAN 50)	192.168.5.0 – 192.168.5.254	/24	3
Users (VLAN 60)	192.168.6.0 – 192.168.6.254	/24	3
Native VLAN	192.168.80.0 – 192.168.80.254	/24	1
Servers (SW4)	10.10.50.0 – 10.10.50.254	/24	3

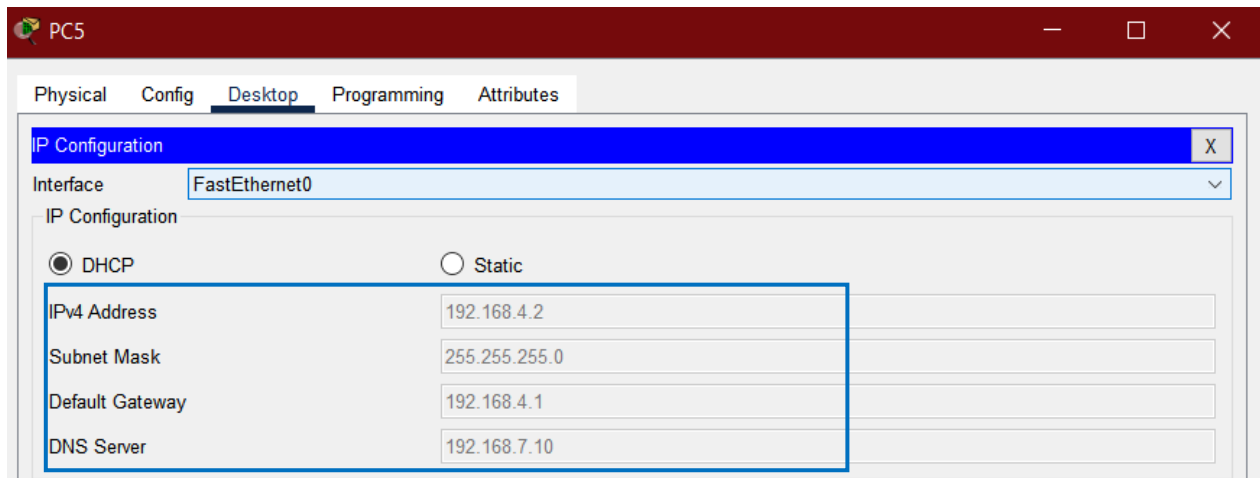
```
Pool HR :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 3
Excluded addresses                : 0
Pending event                     : none

1 subnet is currently in the pool
Current index   IP address range      Leased/Excluded/Total
192.168.3.1    192.168.3.1 - 192.168.3.254  3 / 0 / 254
```

```
Pool IT :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 3
Excluded addresses                : 0
Pending event                     : none

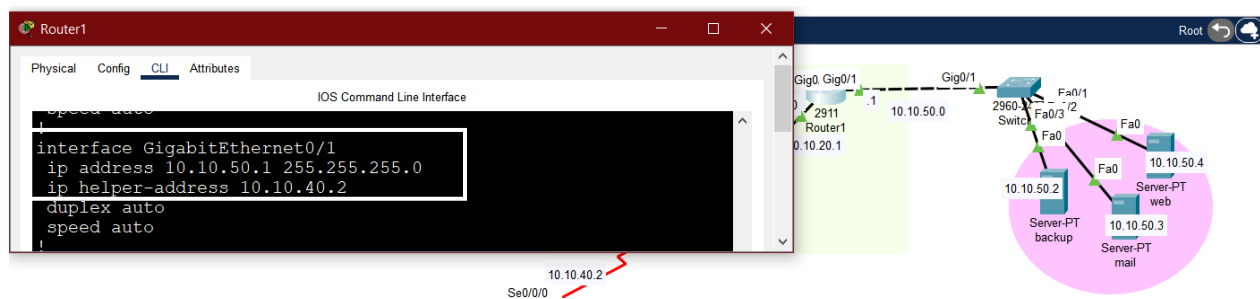
1 subnet is currently in the pool
Current index   IP address range      Leased/Excluded/Total
192.168.1.1    192.168.1.1 - 192.168.1.254  3 / 0 / 254
```

DHCP verification



DHCP relay agent configuration

DHCP relay is configured on router using the ip helper-address command to forward client requests to the DHCP server:



10. AAA (Authentication, Authorization, Accounting)

AAA (Authentication, Authorization, Accounting) is a security framework used to control and monitor access to network resources. It is natively supported on networking platforms from Cisco and commonly integrated with centralized identity solutions like RADIUS or TACACS+.

AAA server configuration

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA**
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

AAA

Service ☒ On ☐ Off Radius Port

Network Configuration

Client Name Client IP

Secret ServerType **Radius**

	Client Name	Client IP	Server Type	Key	
1	Abdo-R	10.10.20.1	Radius	123	Add Save Remove
2	Router3	10.10.20.2	Radius	123	
3	Abdo_R2	10.10.30.1	Radius	123	
4	R3	10.10.40.1	Radius	123	
5	sw4	10.10.50.100	Radius	123	

User Setup

Username Password

	Username	Password	
1	abdo	abdkhaels	Add Save Remove
2	abdosw1	abdkhaels	
3	router3	abdkhaels	

```
radius server 192.168.7.20
  address ipv4 192.168.7.20 auth-port 1645
  key 123
!
!
!
line con 0
  password 7 08204E4A021104121E18
!
line aux 0
!
line vty 0 4
  password 7 08204E4A021104121E18
  login authentication default
  transport input ssh
```

11. SSH Access Configuration

The network is configured to prioritize centralized security while maintaining a local "safety net." This is achieved through the following logic:

- **Primary Method (RADIUS):** When an administrator attempts to SSH into any router or switch, the device first queries the **AAA (RADIUS) Server** (192.168.7.20) for the credentials stored there (e.g., username abdo).
- **Secondary Method (Local):** If the AAA Server is unreachable (offline or network failure), the device automatically falls back to its **Local Database**.

Verification of Fallback Access:

The implementation ensures 100% uptime for administrative access under two scenarios:

Scenario A: AAA Server is Online (Standard Operation)

- **Username:** abdo (or any user defined in the AAA server).
- **Password:** abdkhaels.

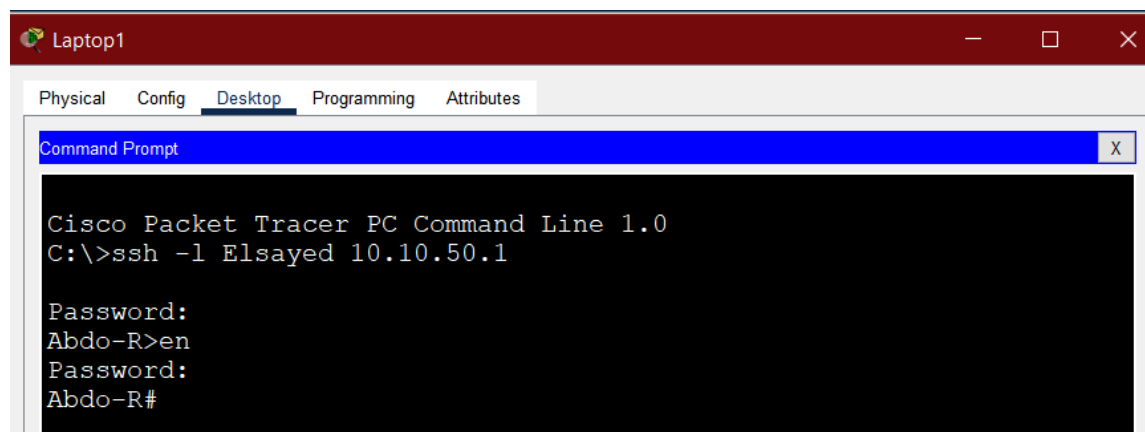


```
C:\>
C:\>
C:\>ssh -l abdo 10.10.30.2

Password:
Abdo_R3>en
Password:
Abdo_R3#
```

Scenario B: AAA Server is Offline (Emergency Access)

- **Username:** Elsayed.
- **Password:** 123.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Elsayed 10.10.50.1

Password:
Abdo-R>en
Password:
Abdo-R#
```

12. DNS Server

The DNS Server (192.168.7.10) acts as the central directory for the network, mapping hostnames to IP addresses to simplify user access.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

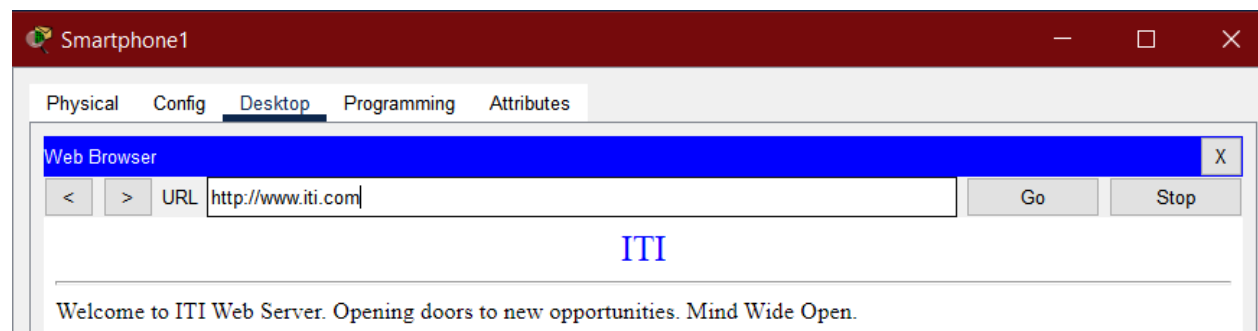
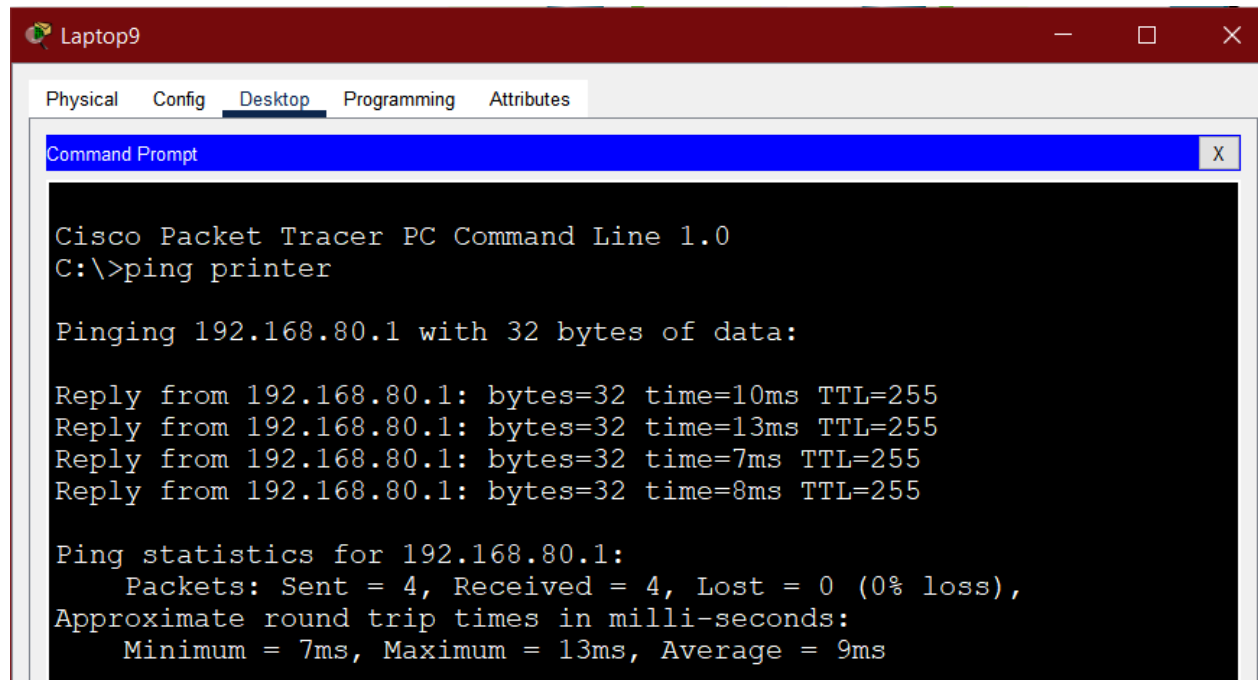
Resource Records

Name Type **A Record** ▼

Address

No.	Name	Type	Detail
0	iti.com	A Record	192.168.7.10
1	lab0	A Record	192.168.3.2
2	lab1	A Record	192.168.3.3
3	lab2	A Record	192.168.3.4
4	lab3	A Record	192.168.2.4
5	lab4	A Record	192.168.2.2
6	lab5	A Record	192.168.2.3
7	lab9	A Record	192.168.6.2
8	pc0	A Record	192.168.1.2
9	pc1	A Record	192.168.1.4
10	pc2	A Record	192.168.1.3
11	phone0	A Record	192.168.6.3
12	phone1	A Record	192.168.6.4
13	printer	A Record	192.168.80.1
14	www.iti.com	A Record	10.10.50.4

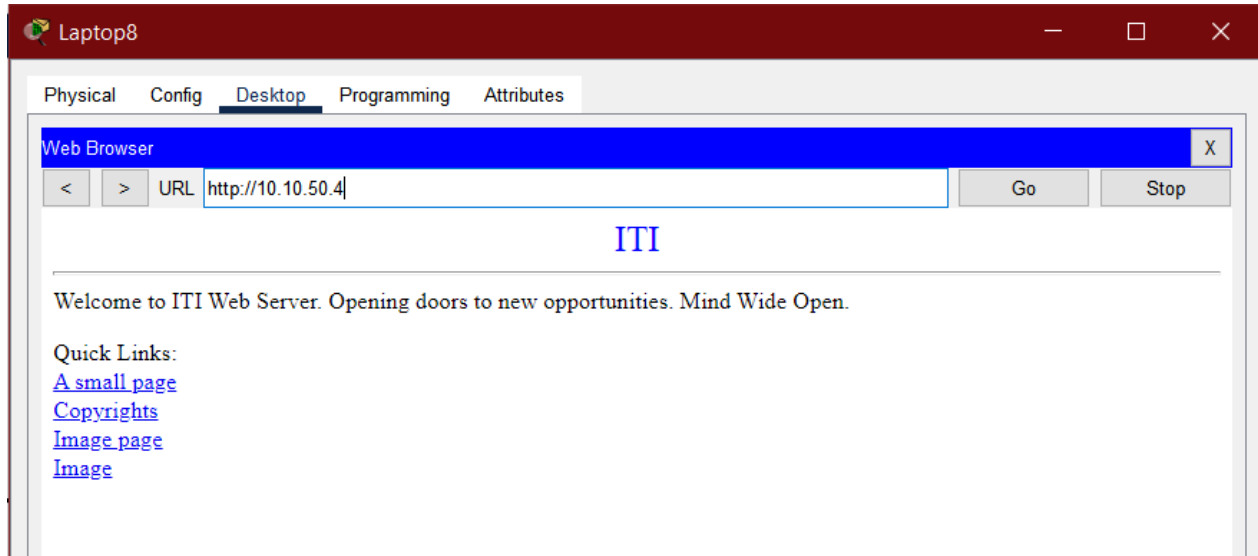
DNS Verification:



13. Web Server

A web server is a dedicated system that hosts and delivers web content and web-based services to clients over the network using the HTTP/HTTPS application protocol.

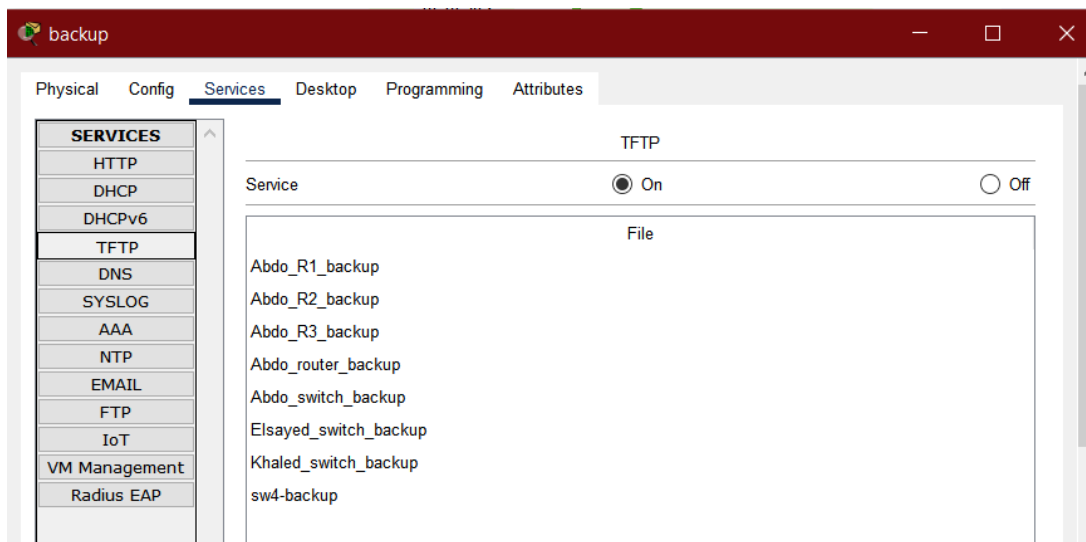
Web server Verification:



14. TFTP Server

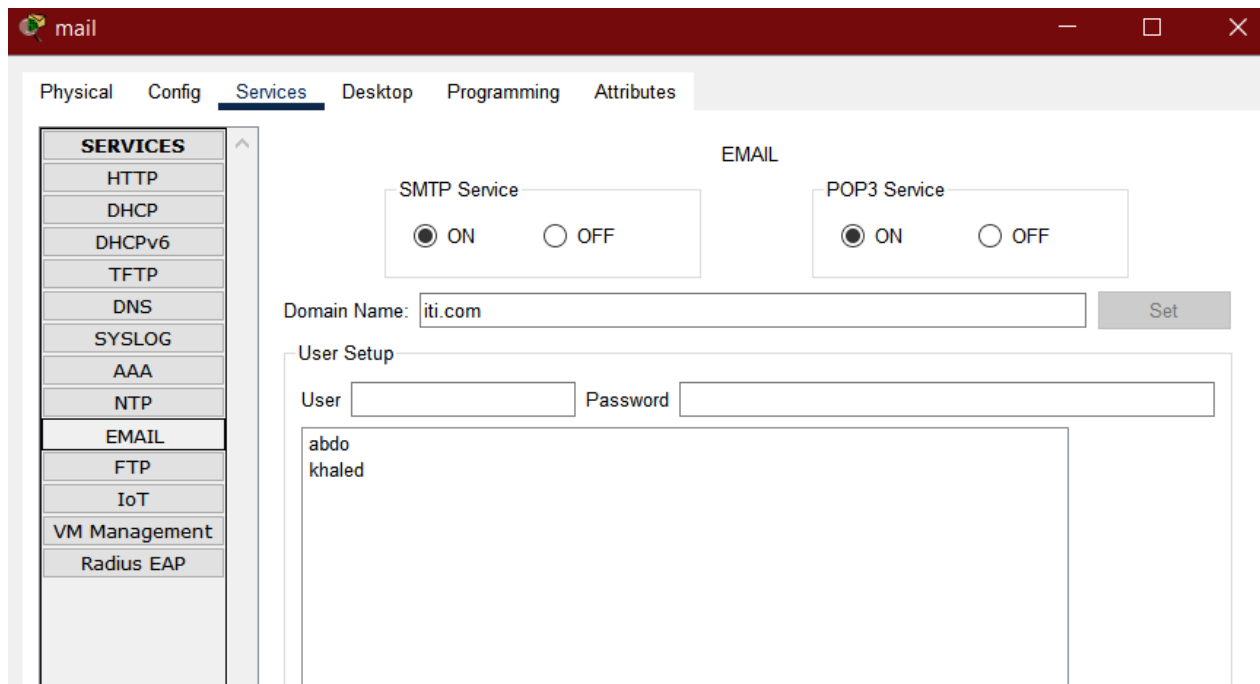
The Backup server (10.10.50.2) acts as a central repository for device configurations.

TFTP server Verification:



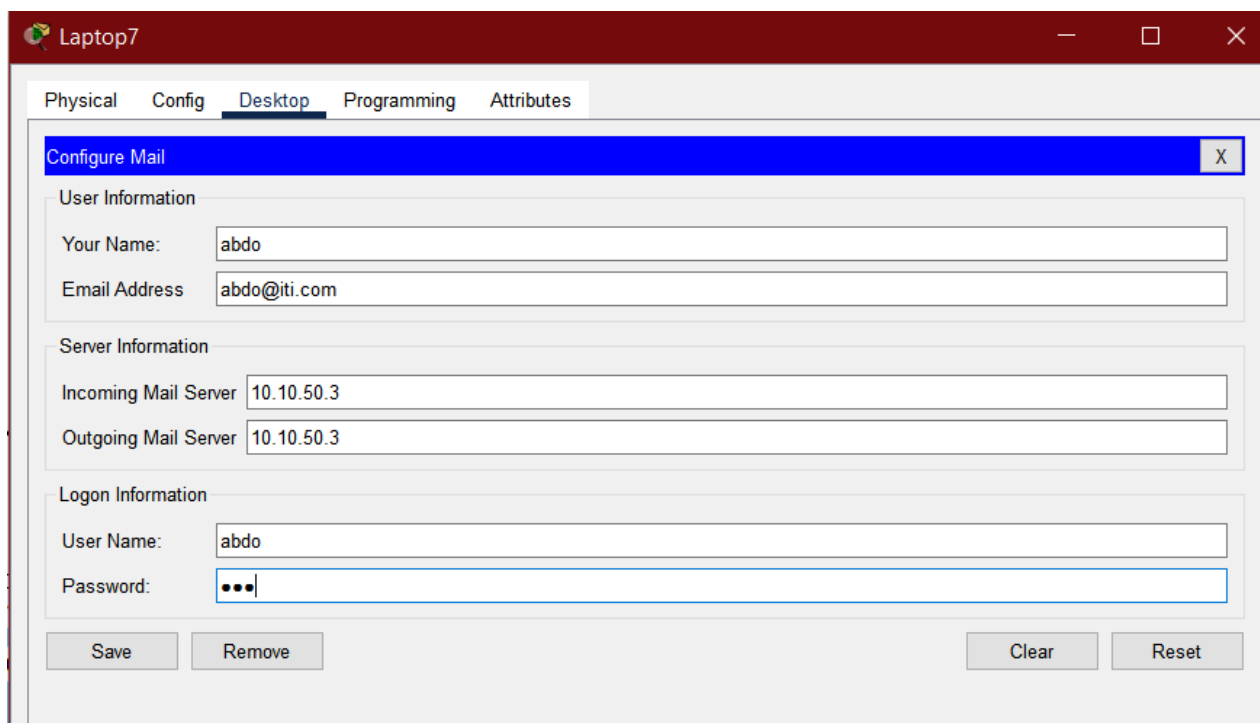
15. Email Server

The enterprise network utilizes a centralized **Mail Server** located at IP address **10.10.50.3** to manage organizational communications. To ensure secure access, user accounts such as **"abdo"** and **"khaled"** have been established with specific credentials.

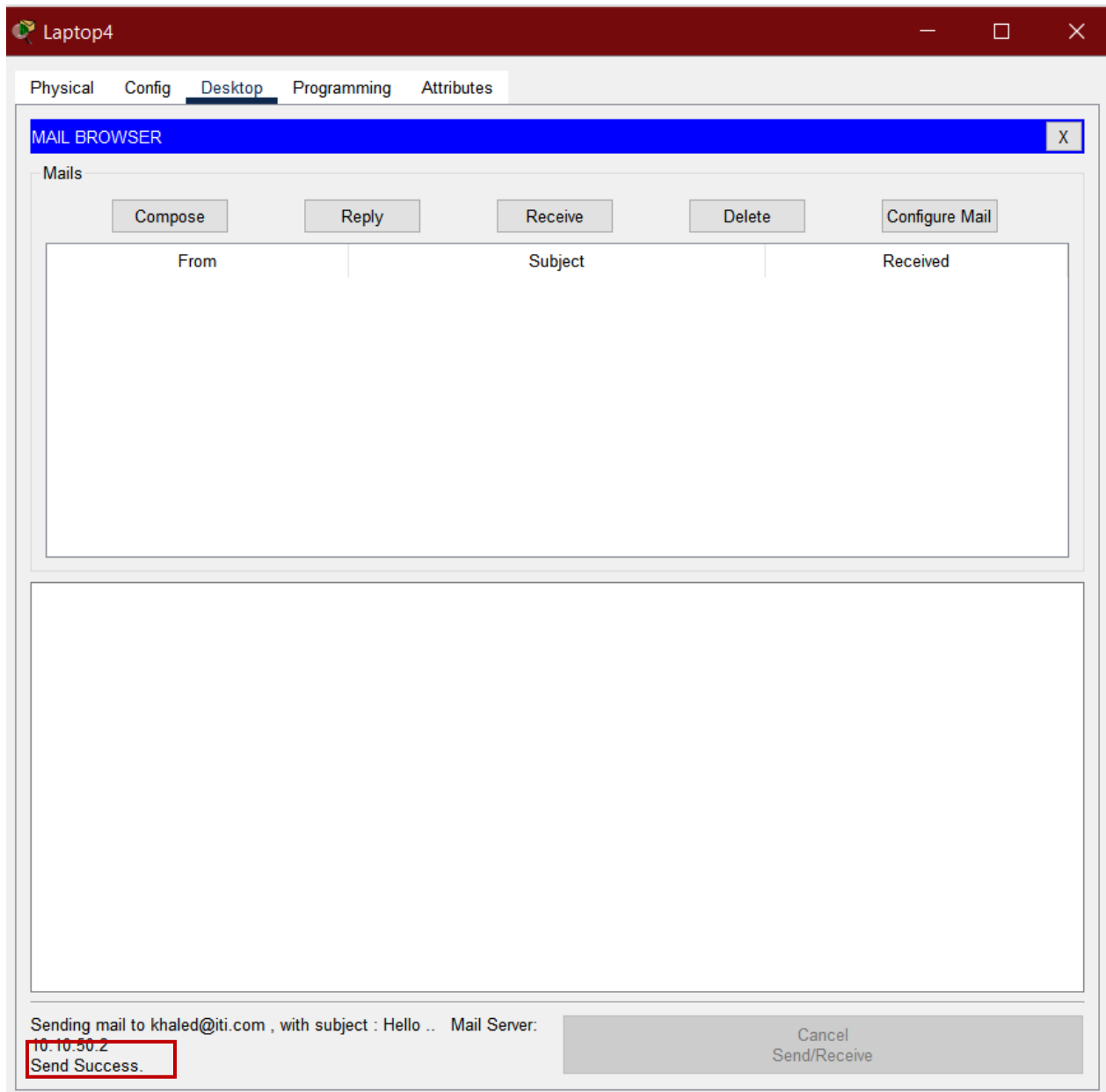


The screenshot shows the 'mail' configuration window with the 'Services' tab selected. On the left, a 'SERVICES' list includes HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL (highlighted), FTP, IoT, VM Management, and Radius EAP. The main area is titled 'EMAIL' and contains two service configuration boxes: 'SMTP Service' and 'POP3 Service', both with 'ON' selected. Below these is a 'Domain Name' field set to 'iti.com' with a 'Set' button. A 'User Setup' section includes 'User' and 'Password' input fields, and a list of users: 'abdo' and 'khaled'.

Email server Verification:



The screenshot shows the 'Laptop7' configuration window with the 'Desktop' tab selected. A 'Configure Mail' dialog box is open, containing three sections: 'User Information' with 'Your Name' (abdo) and 'Email Address' (abdo@iti.com); 'Server Information' with 'Incoming Mail Server' (10.10.50.3) and 'Outgoing Mail Server' (10.10.50.3); and 'Logon Information' with 'User Name' (abdo) and 'Password' (masked with dots). At the bottom are buttons for 'Save', 'Remove', 'Clear', and 'Reset'.



16. Access List

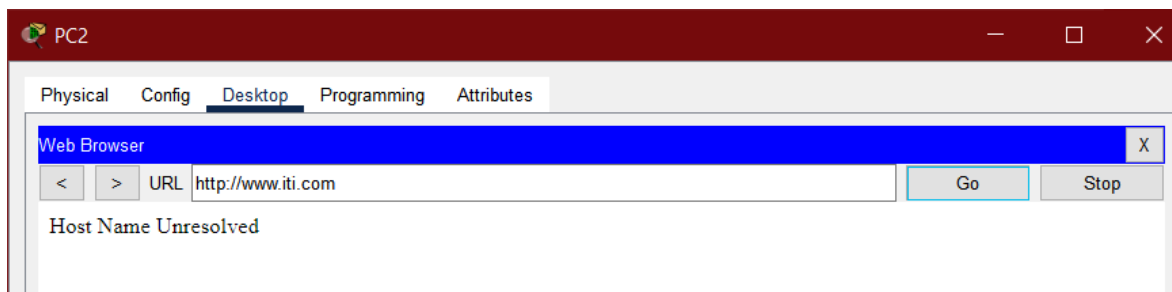
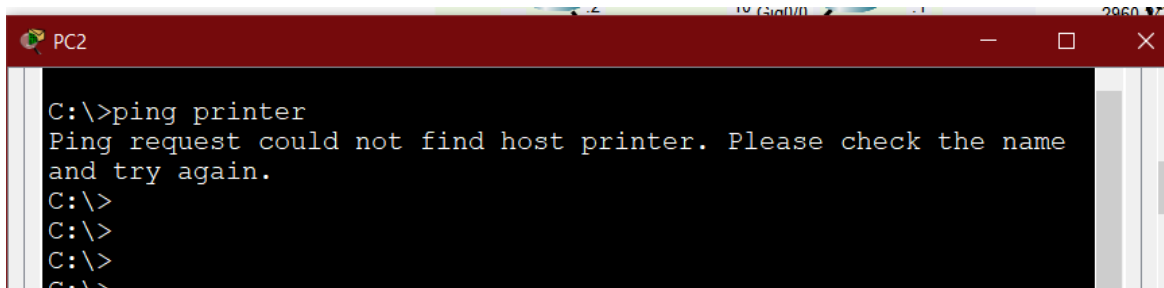
To secure the network and control traffic flow, Extended Access Control Lists (ACLs) were implemented on the routing infrastructure. These rules ensure that critical services like DNS, Web, and Mail are only accessible by authorized hosts and networks.

```
Abdo#show access-lists
Extended IP access list 100
 10 deny tcp 192.168.3.0 0.0.0.255 host 10.10.50.4 eq www (12 match(es))
 20 deny tcp 192.168.3.0 0.0.0.255 host 10.10.50.4 eq 443
 30 permit ip any any (159 match(es))
Extended IP access list 104
 10 deny tcp 192.168.1.0 0.0.0.255 host 10.10.50.3 eq smtp (12 match(es))
 20 deny tcp 192.168.1.0 0.0.0.255 host 10.10.50.3 eq pop3
 30 permit ip any any (16 match(es))
Extended IP access list 102
 10 deny udp host 192.168.1.2 any eq domain
 20 permit ip any any
```

1. DNS Service Restriction (Host Level)

A specific security rule was applied to prevent a single host from performing name resolution queries.

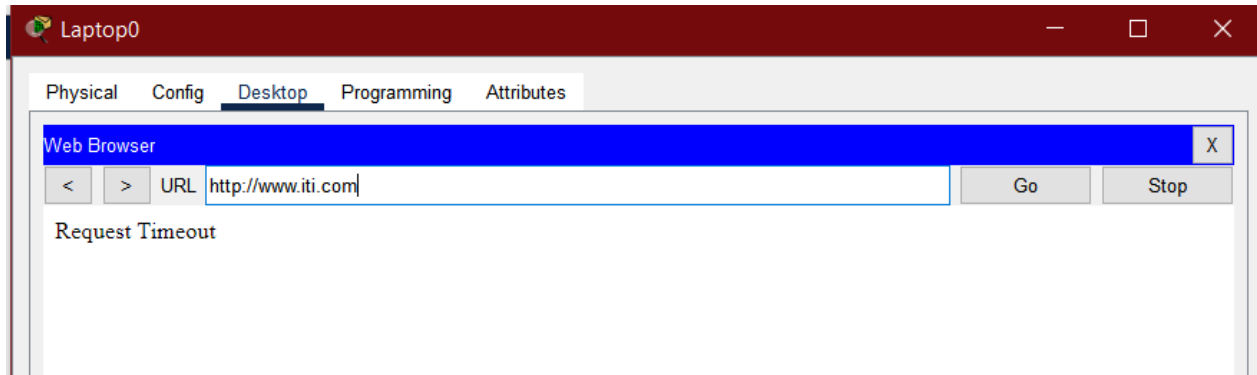
- **Target Host:** 192.168.1.2 (VLAN 10 IT Department).
- **Action:** Deny UDP traffic on port 53 (DNS).
- **Objective:** To restrict this specific host from using the central DNS server (192.168.7.10) while allowing all other network devices to continue name resolution.



2. Web Server Access Control (Network Level)

Access to the internal organization website was restricted for the HR and Wireless department to optimize bandwidth and security.

- **Source Network:** 192.168.3.0/24 (VLAN 30).
- **Destination:** Web Server (10.10.50.4).
- **Action:** Deny TCP traffic on port 80 (HTTP) and port 443 (HTTPS).
- **Objective:** To block the entire HR/Wireless segment from accessing the web server while maintaining their connectivity to other network resources.



3. Mail Server Security (VLAN Level)

A dedicated **Mail Server** was integrated at IP 10.10.50.3 to handle **SMTP** and **POP3** services. A security policy was then enforced to isolate VLAN 10 from email communication.

- **Blocked Segment:** VLAN 10 (192.168.1.0/24).
- **Service Protocols:** SMTP (Port 25) and POP3 (Port 110).
- **Implementation:** An ACL was applied to the router interface to drop any mail-related packets originating from the IT department (VLAN 10).
- **Objective:** To prevent VLAN 10 users from sending or receiving emails via the central mail server for departmental security compliance.

